c. Let $\mu \in S_{c,d}$. Then we claim that $S_{c,d}$ is the coset $\mu S_{c,c}$ of $S_{c,c}$ in S_A. It is obvious that $\mu S_{c,c} \subseteq S_{c,d}$. Let $\sigma \in S_{c,d}$. Then $(\mu^{-1}\sigma)(c) = \mu^{-1}(\sigma(c)) = \mu^{-1}(d) = c$. Thus $\mu^{-1}\sigma \in S_{c,c}$ so $\sigma \in \mu S_{c,c}$ which means that $S_{c,d} \subseteq \mu S_{c,c}$. Hence $S_{c,d} = \mu S_{c,c}$.

45. We can work with $\mathbb{Z}_n$. Let d divide n. Then $\langle n/d \rangle = \{0, n/d, 2n/d, \cdots, (d-1)n/d\}$ is a subgroup of $\mathbb{Z}_n$ of order d. It consists precisely of all elements $x \in \mathbb{Z}_n$ such that $dx = x + x + \cdots + x$ for d summands is equal to 0. Because an element x of any subgroup of order d of $\mathbb{Z}_n$ must satisfy $dx = 0$, we see that $\langle n/d \rangle$ is the only such subgroup. Because the order of a subgroup must divide the order of the whole group, we see that these are the only subgroups that $\mathbb{Z}_n$ has.

46. Every element in $\mathbb{Z}_n$ generates a subgroup of some order d dividing n, and the number of generators of that subgroup is $\varphi(d)$ by Corollary 6.16. By the preceding exercise, there is a unique such subgroup of order d dividing n. Thus $\Sigma_{d|n}\varphi(d)$ counts each element of $\mathbb{Z}_n$ once and only once as a generator of a subgroup of order d dividing n. Hence $\Sigma_{d|n}\varphi(d) = n$.

47. Let d be a divisor of $n = |G|$. Now if G contains a subgroup of order d, then each element of that subgroup satisfies the equation $x^d = e$. By the hypothesis that $x^m = e$ has at most m solutions in G, we see that there can be at most one subgroup of each order d dividing n. Now each $a \in G$ has some order d dividing n, and $\langle a \rangle$ has exactly $\varphi(d)$ generators. Because $\langle a \rangle$ must be the only subgroup of order d, we see that the number of elements of order d for each divisor d of n cannot exceed $\varphi(d)$. Thus we have

$$n = \sum_{d|n} (\text{number of elements of } G \text{ of order } d) \leq \sum_{d|n} \varphi(d) = n.$$

This shows that G must have exactly $\varphi(d)$ elements of each order d dividing n, and thus must have $\varphi(n) \geq 1$ elements of order n. Hence G is cyclic.

11. Direct Products and Finitely Generated Abelian Groups

1. (See the answer in the text.)

2. The group is cyclic because there are elements of order 12.

Element	Order	Element	Order	Element	Order
(0,0)	1	(1,0)	3	(2,0)	3
(0,1)	4	(1,1)	12	(2,1)	12
(0,2)	2	(1,2)	6	(2,2)	6
(0,3)	4	(1,3)	12	(2,3)	12

3. lcm(2, 2) = 2. (The abbreviation lcm stands for *least common multiple*.)

4. lcm(3, 5) = 15. (The abbreviation lcm stands for *least common multiple*.)

5. lcm(3, 9) = 9. (The abbreviation lcm stands for *least common multiple*.)

6. lcm(4, 6, 5) = 60. (The abbreviation lcm stands for *least common multiple*.)

7. lcm(4, 2, 5, 3) = 60. (The abbreviation lcm stands for *least common multiple*.)

8. For $\mathbb{Z}_6 \times \mathbb{Z}_8$: the lcm (*least common multiple*) of 6 and 8 which is 24.
For $\mathbb{Z}_{12} \times \mathbb{Z}_{15}$: the lcm of 12 and 15 which is 60.

9. $\{(0,0),(1,0)\}$ $\{(0,0),(0,1)\}$ $\{(0,0),(1,1)\}$

10. There are 7 order 2 subgroups: $\langle(1,0,0)\rangle, \langle(0,1,0)\rangle, \langle(0,0,1)\rangle, \langle(1,1,0)\rangle, \langle(1,0,1)\rangle, \langle(0,1,1)\rangle, \langle(1,1,1)\rangle$.

There are 7 order 4 subgroups:
$\{(0,0,0),(1,0,0),(0,1,0),(1,1,0)\}$ $\{(0,0,0),(1,0,0),(0,0,1),(1,0,1)\}$
$\{(0,0,0),(1,0,0),(0,1,1),(1,1,1)\}$ $\{(0,0,0),(1,1,0),(0,0,1),(1,1,1)\}$
$\{(0,0,0),(1,1,0),(0,1,1),(1,0,1)\}$ $\{(0,0,0),(1,1,1),(0,1,0),(1,0,1)\}$
$\{(0,0,0),(0,1,1),(0,0,1),(0,1,0)\}$

11. (See the answer in the text.)

12. $\{(0,0,0),(1,0,0),(0,1,0),(1,1,0)\}$ $\{(0,0,0),(1,0,0),(0,0,2),(1,0,2)\}$
$\{(0,0,0),(1,0,0),(0,1,2),(1,1,2)\}$ $\{(0,0,0),(1,1,0),(0,0,2),(1,1,2)\}$
$\{(0,0,0),(1,1,0),(0,1,2),(1,0,2)\}$ $\{(0,0,0),(1,1,2),(0,1,0),(1,0,2)\}$
$\{(0,0,0),(0,1,2),(0,0,2),(0,1,0)\}$

13. $\mathbb{Z}_3 \times \mathbb{Z}_{20}$, $\mathbb{Z}_4 \times \mathbb{Z}_{15}$, $\mathbb{Z}_5 \times \mathbb{Z}_{12}$, $\mathbb{Z}_3 \times \mathbb{Z}_4 \times \mathbb{Z}_5$

14. **a.** 4 **b.** 12 **c.** 12 **d.** 2, 2 **e.** 8

15. The maximum possible order is $12 = \text{lcm}(4, 6)$.

16. Yes. Both groups are isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_4$.

17. The maximum possible order is $120 = \text{lcm}(8, 20, 24)$.

18. No. $\mathbb{Z}_8 \times \mathbb{Z}_{10} \times \mathbb{Z}_{24} \simeq \mathbb{Z}_2 \times \mathbb{Z}_8 \times \mathbb{Z}_8 \times \mathbb{Z}_3 \times \mathbb{Z}_5$ but $\mathbb{Z}_4 \times \mathbb{Z}_{12} \times \mathbb{Z}_{40} \simeq \mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_8 \times \mathbb{Z}_3 \times \mathbb{Z}_5$

19. The maximum possible order is $180 = \text{lcm}(4, 18, 15)$.

20. Yes. Both groups are isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_9 \times \mathbb{Z}_5$.

21. $\mathbb{Z}_8$, $\mathbb{Z}_2 \times \mathbb{Z}_4$, $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$

22. $\mathbb{Z}_{16}$, $\mathbb{Z}_2 \times \mathbb{Z}_8$, $\mathbb{Z}_4 \times \mathbb{Z}_4$, $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_4$, $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$

23. (See the answer in the text.)

24. $\mathbb{Z}_{16} \times \mathbb{Z}_9 \times \mathbb{Z}_5$, $\mathbb{Z}_2 \times \mathbb{Z}_8 \times \mathbb{Z}_9 \times \mathbb{Z}_5$, $\mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_9 \times \mathbb{Z}_5$,
$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_9 \times \mathbb{Z}_5$, $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_9 \times \mathbb{Z}_5$,
$\mathbb{Z}_{16} \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$, $\mathbb{Z}_2 \times \mathbb{Z}_8 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$,
$\mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$, $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$,
$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$

25. (See the answer in the text.)

26. There are 3 of order 24, arising from the subscript sequences 8, 3 and 2, 4, 3 and 2, 2, 2, 3 on the factors $\mathbb{Z}$. Similarly, there are 2 of order 25 arising from the subscript sequences 25 and 5, 5. There are $3 \cdot 2 = 6$ of order $24 \cdot 25$, because each of three for order 24 can be paired with each of the two of order 25.

27. Because there are no primes that divide both m and n, any abelian group of order mn is isomorphic to a direct product of cyclic groups of prime-power order where all cyclic groups given by primes dividing m appear before any of the primes dividing n. Thus any abelian group of order mn is isomorphic to a direct product of a group of order m with a group of order n, when $\gcd(m, n) = 1$. Because there are r choices for the group of order m and s choices for the group of order n, there are rs choices in all.

28. We have $10^5 = 2^5 5^5$. There are 7 groups of order 2^5, up to isomorphism, by Exercise 23. Replacing factors 2 by factors 5 in the answer to Exercise 23, we see that there are also 7 abelian groups of order 5^5, up to isomorphism. By Exercise 27, there are $7 \cdot 7 = 49$ abelian groups of order 10^5, up to isomorphism.

29. **a.** We just illustrate with the computation for groups of order p^8, to get the last entry 22 in the table. We try to be systematic, according as there is just one factor $\mathbb{Z}$, then two factors $\mathbb{Z}$, then three, etc. For each of these cases, we list the possible sequences of exponents i that appear on the subscripts p^i on the factors $\mathbb{Z}_{p^i}$.

Factors	Exponent Sequences	Total
1	8	1
2	$1,7 \quad 2,6 \quad 3,5 \quad 4,4$	4
3	$1,1,6 \quad 1,2,5 \quad 1,3,4 \quad 2,2,4 \quad 2,3,3$	5
4	$1,1,1,5 \quad 1,1,2,4 \quad 1,1,3,3 \quad 1,2,2,3 \quad 2,2,2,2$	5
5	$1,1,1,1,4 \quad 1,1,1,2,3 \quad 1,1,2,2,2$	3
6	$1,1,1,1,1,3 \quad 1,1,1,1,2,2$	2
7	1,1,1,1,1,1,2	1
8	1,1,1,1,1,1,1,1	1

Thus there are a total of $1 + 4 + 5 + 5 + 3 + 2 + 1 + 1 = 22$ abelian groups of order p^8, up to isomorphism.

b. We use the entries from the table in the answer in the text.
i) $3 \cdot 5 \cdot 15 = 225$ *ii*) $5 \cdot 15 = 225$ *iii*) $q^5 r^4 q^3 = q^8 r^4$ so our computation becomes $22 \cdot 5 = 110$

30. Finish this diagram by a double arrow at the right end of each row looping around to the left end of the row, and a single arrow at the bottom of each column looping around to the top of the column. LaTeX can't do dashed arrows or the looping ones.

$$
\begin{array}{ccccccccc}
(0,0) & \Rightarrow & (0,1) & \Rightarrow & (0,2) & \Rightarrow & \cdots & \Rightarrow & (0,m) \\
\downarrow & & \downarrow & & \downarrow & & \cdots & & \downarrow \\
(1,0) & \Rightarrow & (1,1) & \Rightarrow & (1,2) & \Rightarrow & \cdots & \Rightarrow & (1,m) \\
\downarrow & & \downarrow & & \downarrow & & \cdots & & \downarrow \\
(2,0) & \Rightarrow & (2,1) & \Rightarrow & (2,2) & \Rightarrow & \cdots & \Rightarrow & (2,m) \\
\downarrow & & \downarrow & & \downarrow & & \cdots & & \downarrow \\
\vdots & & \vdots & & \vdots & & \vdots & & \vdots \\
\downarrow & & \downarrow & & \downarrow & & \cdots & & \downarrow \\
(n,0) & \Rightarrow & (n,1) & \Rightarrow & (n,2) & \Rightarrow & \cdots & \Rightarrow & (n,m)
\end{array}
$$

$$
\begin{array}{cc}
(0,1) & (1,0) \\
\Longrightarrow & \longrightarrow
\end{array}
$$

31. **a.** It is abelian if the two generators a and b representing the two arc types commute. From a diagram, we check that this is the case when the arrows on both n-gons have the same (clockwise or counterclockwise) direction.
b. $\mathbb{Z}_2 \times \mathbb{Z}_n$ **c.** $\mathbb{Z}_2 \times \mathbb{Z}_n$ is cyclic when n is odd.
d. It is isomorphic to the dihedral group D_n, for it is generated by an element ρ (a rotation) of order n and an element μ (a reflection) of order 2 satisfying $\rho\mu = \mu\rho^{-1}$.

32. T T F T F F F F F T **33.** $\mathbb{Z}_p$ is an example for any prime p.

34. **a.** Cardinality considerations show that the only subgroup of $\mathbb{Z}_5 \times \mathbb{Z}_6$ that it isomorphic to $\mathbb{Z}_5 \times \mathbb{Z}_6$ is $\mathbb{Z}_5 \times \mathbb{Z}_6$ itself.

b. There are an infinite number of them. Subgroup $m\mathbb{Z} \times n\mathbb{Z}$ is isomorphic to $\mathbb{Z} \times \mathbb{Z}$ for all positive integers m and n.

35. S_3 is an example, for its nontrivial proper subgroups are all abelian, so any direct product of them would be abelian, and could not be isomorphic to nonabelian S_3.

36. T F F T T F T F T T **37.** The numbers are the same.

38. **a.** Yes, it has just one subgroup of order 8 because $72 = 8 \cdot 9$ so the subgroup of order 8 consists of all elements having order that divides 8.

b. No. If the group is $\mathbb{Z}_8 \times \mathbb{Z}_9$, then it has just one subgroup $\{(0,0),(2,0),(4,0),(6,0)\}$ of order 4, but if it is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_9$, it has more than one subgroup of order 4, namely

$$\{(0,0,0),(0,1,0),(0,2,0),(0,3,0)\} \text{ and } \{(0,0,0),(2,0,0),(0,2,0),(2,2,0)\}.$$

39. Let G be abelian and let $a, b \in G$ have finite order. Then $a^r = b^s = e$ for some positive integers r and s. Because G is abelian, we see that $(ab)^{rs} = (a^r)^s(b^s)^r = e^s e^r = ee = e$, so ab has finite order. This shows that the subset H of G consisting of all elements of finite order is closed under the group operation. Of course $e \in H$ because e has order 1. If $a^r = e$, then $a^{-r} = (a^{-1})^r = e$ also, showing that $a \in H$ implies $a^{-1} \in H$, and completing the demonstration that H is a subgroup of G.

40. The torsion subgroup of $\mathbb{Z}_4 \times \mathbb{Z} \times \mathbb{Z}_3$ has order $4 \cdot 3 = 12$. The torsion subgroup of $\mathbb{Z}_{12} \times \mathbb{Z} \times \mathbb{Z}_{12}$ has order $12 \cdot 12 = 144$.

41. $\{-1, 1\}$ **42.** $\{e^{q\pi i} \mid q \in \mathbb{Q}\}$

43. Let G be a finitely generated abelian group and write it (up to isomorphism) in the form described in Theorem 11.12. Put parentheses around the first portion, involving factors of the form $\mathbb{Z}_{p^r}$, and then put parenthese around the second part, containing the factors $\mathbb{Z}$. We have then exhibited G, up to isomorphism, in the form $H \times K$ where H is a torsion group and K is torsion free.

44. **a.** 36 ; **b.** 2, 12, and 60 as explained in Part(**c**).

c. Find an isomorphic group that is a direct product of cyclic groups of prime-power order. For each prime divisor of the order of the group, write the subscripts in the direct product involving that prime in a row in order of increasing magnitude. Keep the right-hand ends of the rows aligned. Then take the product of the numbers down each column of the array. These are the torsion coefficients. Illustrating with the group in **b.**, we first form $\mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_5$. We now form the array and multiply columns, as in

$$\begin{array}{cccc} 2 & 4 & 4 \\ & 3 & 3 \\ & & 5 \\ \hline 2 & 12 & 60 \end{array}$$

obtaining the torsion coefficients 2, 12, 60.

45. If m and n are relatively prime, then (1, 1) has order mn so the group is cyclic of order mn. If m and n are not relatively prime, then no element has order exceeding the least common multiple of m and n, which has to be less than mn, so the group is not cyclic.

46. Computation in a direct product of n groups consists of computing using the individual group operations in each of the n components. In a direct product of abelian groups, the individual group operations are all commutative, and it follows at once that the direct product is an abelian group.

47. *Closure:* Let $a, b \in H$. Then $a^2 = b^2 = e$. Because G is abelian, we see that $(ab)^2 = abab = aabb = ee = e$, so $ab \in H$ also. Thus H is closed under the group operation.

Identity: We are given that $e \in H$.

Inverses: For all $a \in H$, the equation $a^2 = e$ means that $a^{-1} = a \in H$. Thus H is a subgroup.

48. Yes, H is a subgroup for order 3, by essentially the same proof as in the preceding exercise. No, H is not a subgroup for order 4, because the square of an element of order 4 has order 2, so H is not closed under the operation. For prime positive integers, H will be a subgroup.

49. S_3 is a counterexample.

50. **a.** $(h, k) = (h, e)(e, k)$ **b.** $(h, e)(e, k) = (h, k) = (e, k)(h, e)$. **c.** The only element of $H \times K$ of the form (h, e) and also of the form (e, k) is $(e, e) = e$.

51. *Uniqueness:* Suppose that $g = hk = h_1k_1$ for $h, h_1 \in H$ and $k, k_1 \in K$. Then $h_1^{-1}h = k_1k^{-1}$ is in both H and K, and we know that $H \cap K = \{e\}$. Thus $h_1^{-1}h = k_1k^{-1} = e$, from which we see that $h = h_1$ and $k = k_1$.

Isomorphic: Suppose $g_1 = h_1k_1$ and $g_2 = h_2k_2$. Then $g_1g_2 = h_1k_1h_2k_2 = h_1h_2k_1k_2$ because elements of H and K commute by hypothesis **b.** Thus by uniqueness, g_1g_2 is renamed $(h_1h_2, k_1k_2) = (h_1, k_1)(h_2, k_2)$ in $H \times K$.

52. Recall that every subgroup of a cyclic group is cyclic. Thus if a finite abelian group G contains a subgroup isomorphic to $\mathbb{Z}_p \times \mathbb{Z}_p$, which is not cyclic, then G cannot be cyclic.

Conversely, suppose that G is a finite abelian group that is not cyclic. By Theorem 11.12, G contains a subgroup isomorphic to $\mathbb{Z}_{p^r} \times \mathbb{Z}_{p^s}$ for the same prime p, because if all components in the direct product correspond to distinct primes, then G would be cyclic by Theorem 11.5. The subgroup $\langle p^{r-1} \rangle \times \langle p^{s-1} \rangle$ of $\mathbb{Z}_{p^r} \times \mathbb{Z}_{p^s}$ is clearly isomorphic to $\mathbb{Z}_p \times \mathbb{Z}_p$.

53. By the Theorem of Lagrange, the order of an element of a finite group (that is, the order of the cyclic subgroup it generates) divides the order of the group. Thus if G has prime-power order, then the order of every element is also a power of the prime. The hypothesis of commutativity was not used.

54. By Theorem 11.12, the groups that appear in the decompositions of $G \times K$ and of $H \times K$ are unique except for the order of the factors. Because $G \times K$ and $H \times K$ are isomorphic, these factors in their decompositions must be the same. Because the decompositions of $G \times K$and $H \times K$can both be written in the order with the factors from K last, we see that G and H must have the same factors in their expression in the decomposition described in Theorem 11.12. Thus G and H are isomorphic.